

Nonstandard Machine Learning Algorithms for Microarray Data Mining

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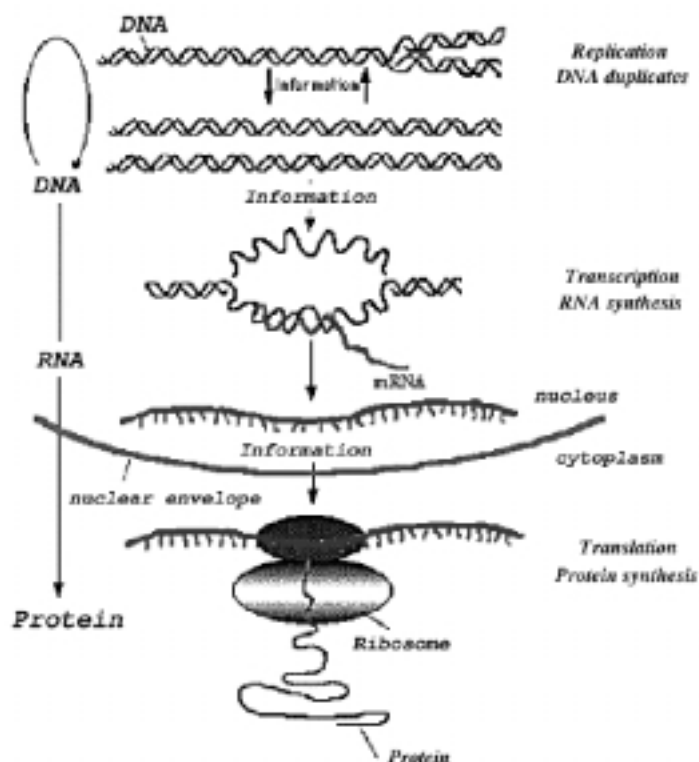
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Outline

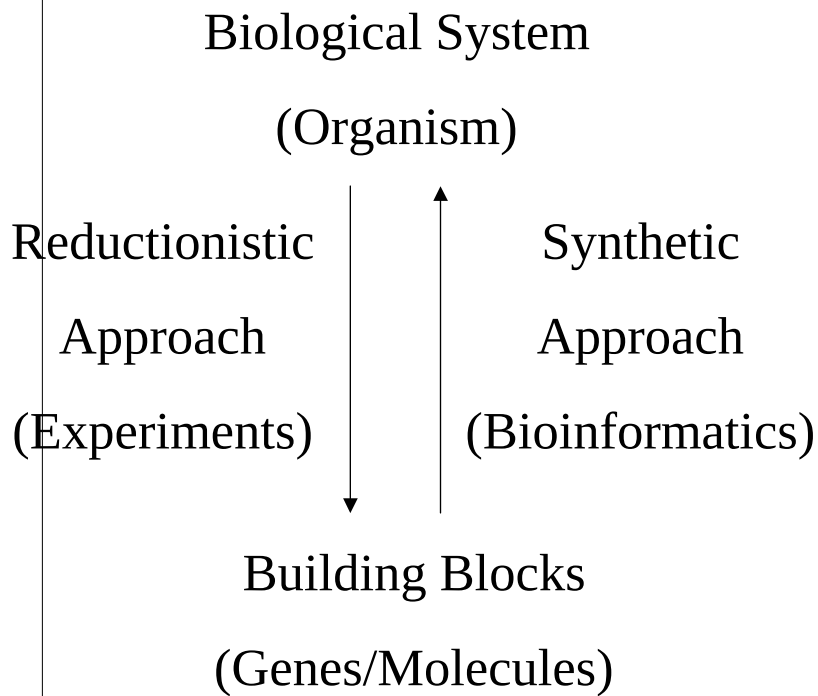
- Microarray Bioinformatics
- Probabilistic Graphical Models for Gene Expression Analysis
- Gene-Drug Dependency Analysis with Bayesian Networks
- Gene and Tissue Clustering with Latent Variable Models
- Summary and Future Work

Microarray Bioinformatics

Molecular Biology: Central Dogma



Reductionistic and Synthetic Approaches in Biology



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Topics in Bioinformatics

Sequence analysis

- ▶ Sequence alignment
- ▶ Structure and function prediction
- ▶ Gene finding

Structure analysis

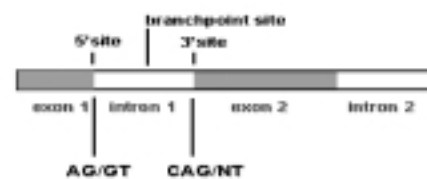
- ▶ Protein structure comparison
- ▶ Protein structure prediction
- ▶ RNA structure modeling

Expression analysis

- ▶ Gene expression analysis
- ▶ Gene clustering

Pathway analysis

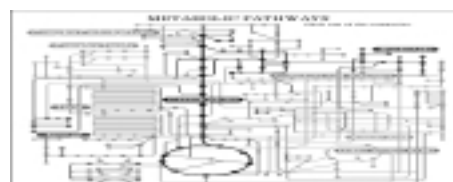
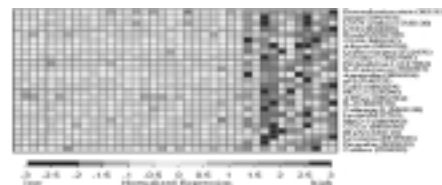
- ▶ Metabolic pathway
- ▶ Regulatory networks



Given a Protein's Primary Structure -- Amino Acid Sequence

...HIS-CYS-ALA-ALA-GLY-GLU-ASP...

Can We Determine Its 3D Structure?

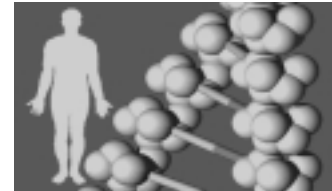
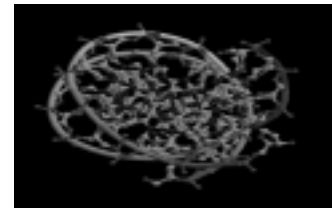


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Applications of DNA Microarrays

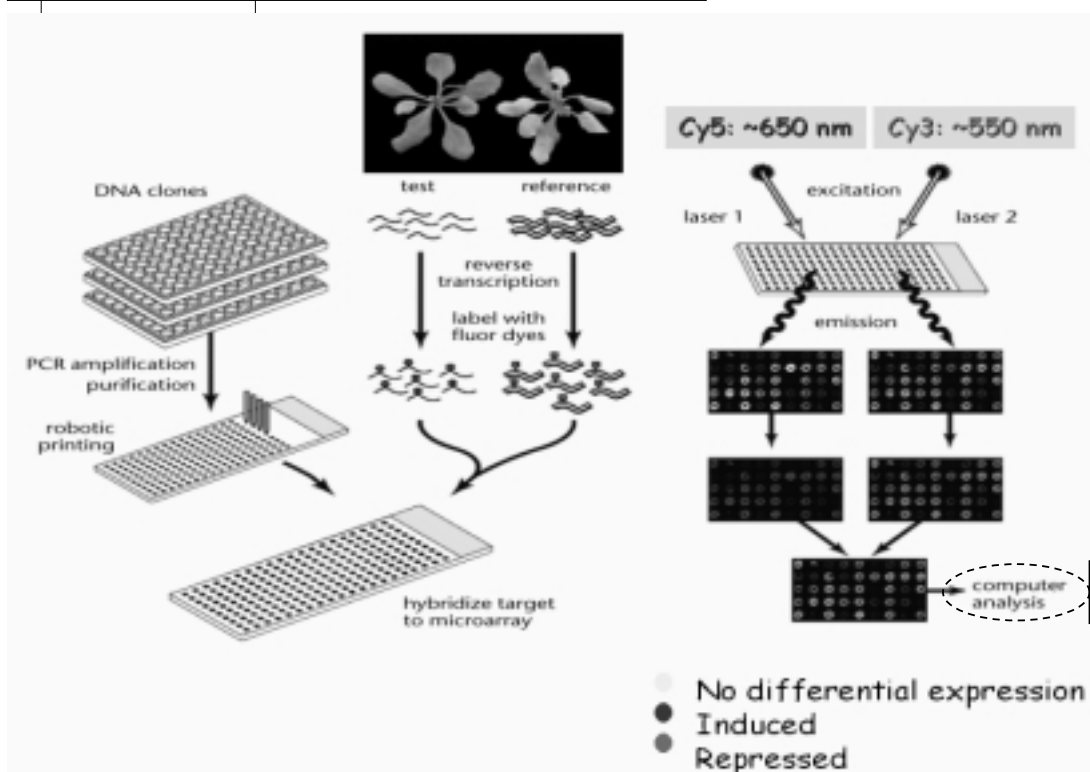
- Gene discovery
- Analysis of gene regulation
- Disease diagnosis
- Drug discovery: pharmacogenomics
- Toxicological research: toxicogenomics



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A Comparative Hybridization Experiment

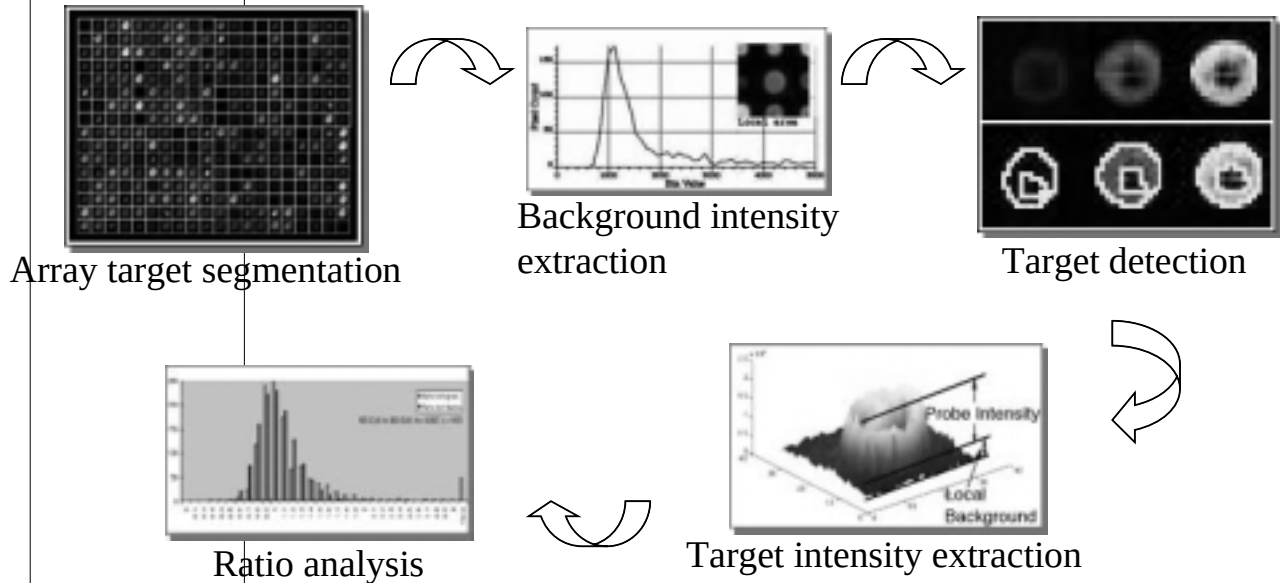


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Image Analysis

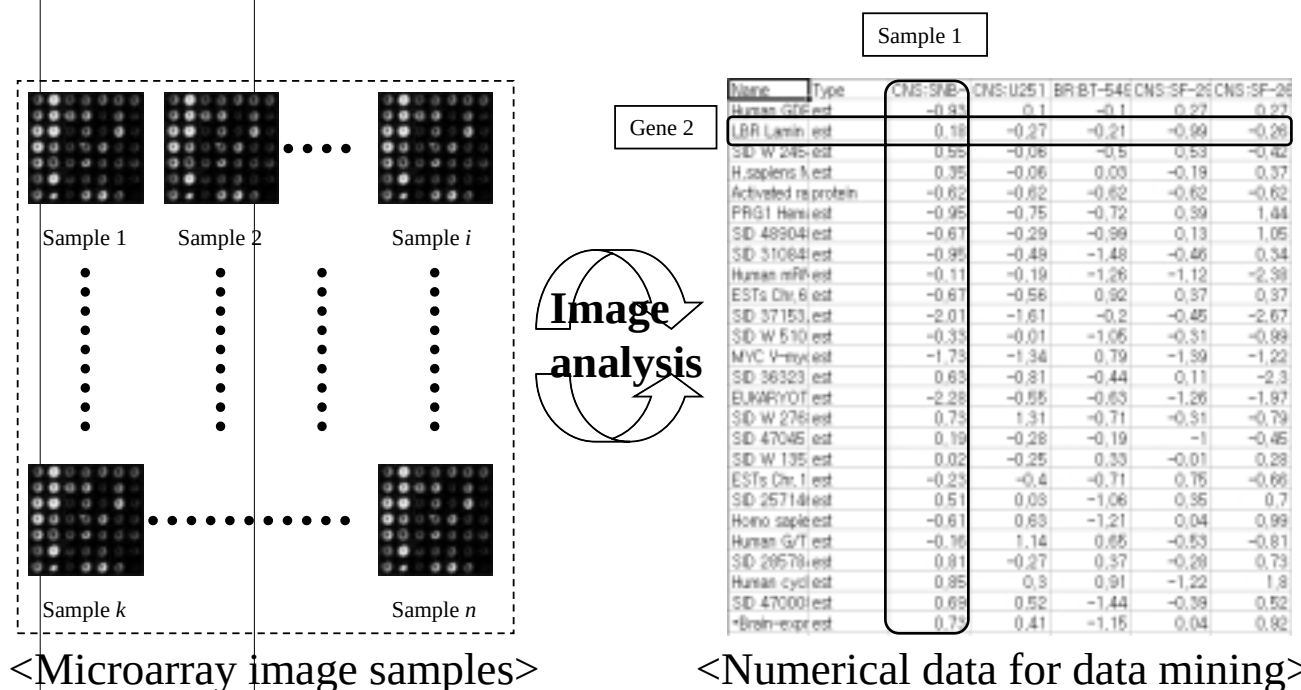
- Scanned images → probe intensities → numerical values for higher-level analysis



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Data Preparation for Data Mining



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Gene Expression Data Mining

Gene	Cell 1	Cell 2	Cell 3	Cell 4	Cell 5	Cell 6
Human 1000	-0.01	0.1	-0.1	0.01	-0.01	0.01
Human 2000	0.01	-0.01	-0.01	-0.01	-0.01	-0.01
Human 3000	0.01	-0.01	-0.01	-0.01	-0.01	-0.01
Human 4000	0.01	-0.01	-0.01	-0.01	-0.01	-0.01
Human 5000	0.01	-0.01	-0.01	-0.01	-0.01	-0.01
Human 6000	0.01	-0.01	-0.01	-0.01	-0.01	-0.01
Human 7000	0.01	-0.01	-0.01	-0.01	-0.01	-0.01
Human 8000	0.01	-0.01	-0.01	-0.01	-0.01	-0.01
Human 9000	0.01	-0.01	-0.01	-0.01	-0.01	-0.01
Human 10000	0.01	-0.01	-0.01	-0.01	-0.01	-0.01

Data preprocessing:

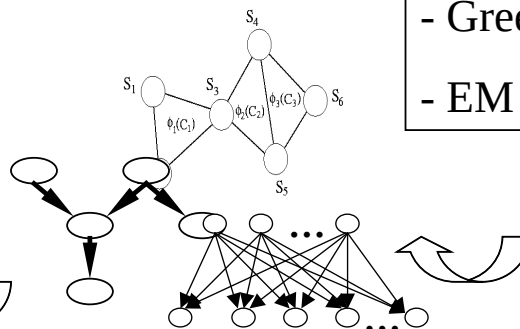
- Normalization
- Discretization
- Gene selection

Mining

- Classification
- Clustering
- Regulation analysis

Learning:

- Greedy search
- EM algorithm

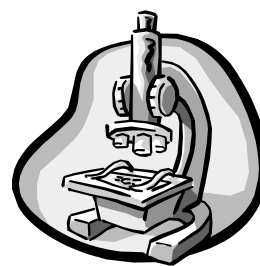


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Why Data Mining?

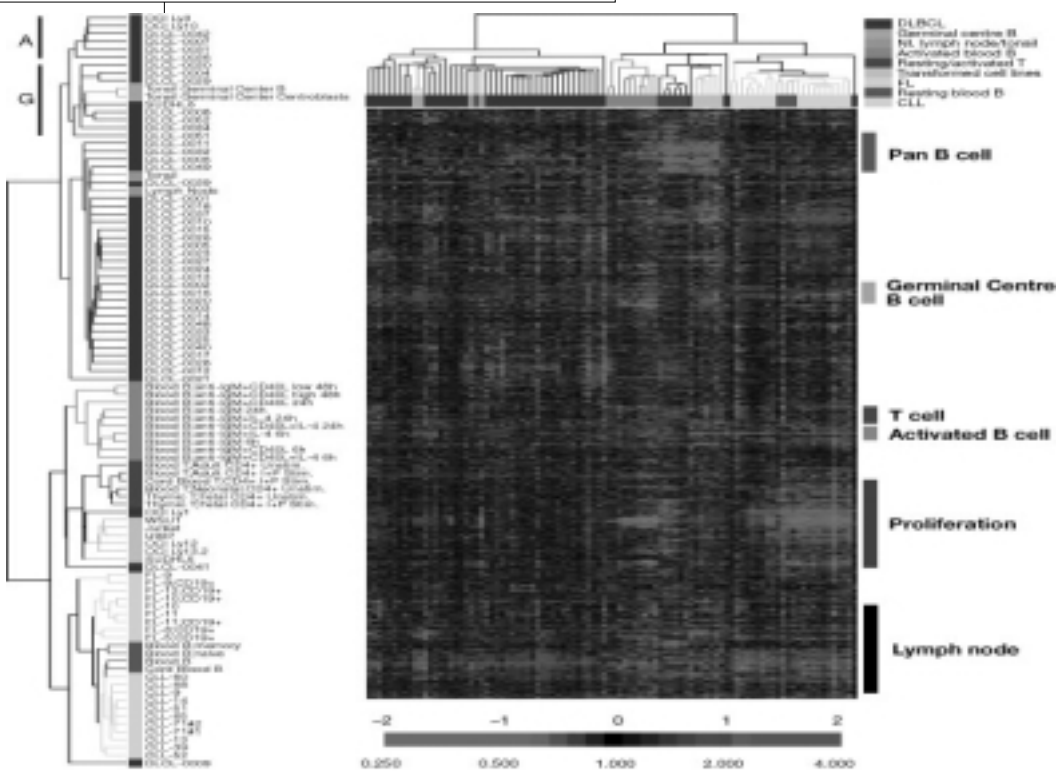
- Traditional analysis
 - ◆ One gene in one experiment
 - ◆ Small data sets
 - ◆ Simple analytical methods
- High-throughput genomic analysis
 - ◆ Simultaneous measurements of thousands of gene expression levels
→ massive data sets
 - ◆ Statistical methods
 - ◆ Machine learning approach



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An Example of Data Mining: Clustering



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Analysis of DNA Microarray Data

Previous Work

- Characteristics of data
 - ◆ Analysis of expression ratio based on each sample
 - ◆ Analysis of time-variant data
- Clustering
 - ◆ Self-organizing maps [Golub et al., 1999]
 - ◆ Singular value decomposition [Orly Alter et al., 2000]
- Classification
 - ◆ Support vector machines [Brown et al., 2000]
- Gene identification
 - ◆ Information theory [Stefanie et al., 2000]
- Gene modeling
 - ◆ Bayesian networks [Friedman et.al., 2000]

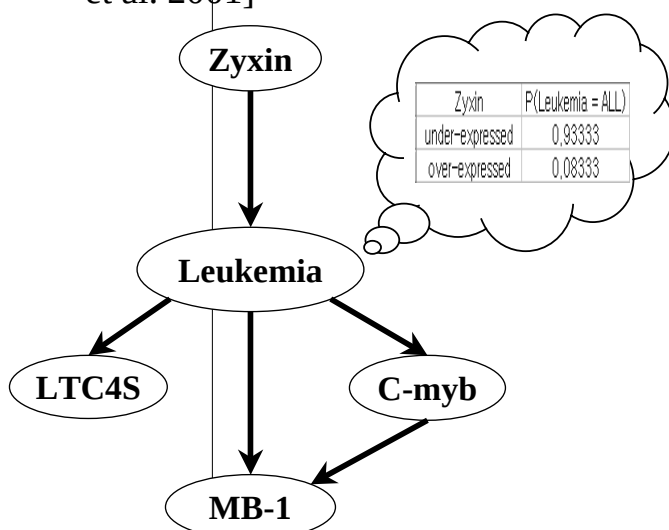
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Probabilistic Graphical Models for Gene Expression Analysis

An Introductory Example

- A Bayesian network classifier for acute leukemias [Hwang et al. 2001]



<Network structure>

Zyxin	Zyxin
Leukemia	ALL or AML
LTC4S	Leukotriene C4 synthase (LTC4S) gene
C-myb	C-myb gene extracted from Human (c-myb) gene, complete primary cds, and five complete alternatively spliced cds
MB-1	MB-1 gene

$$P(\mathbf{X}) = \prod_{i=1}^n P(X_i | \mathbf{Pa}_i)$$

Probabilistic Graphical Models

- The joint probability distribution over $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$

♦ Chain rule
$$P(\mathbf{X}) = \prod_{i=1}^n P(X_i | X_1, \dots, X_{i-1})$$

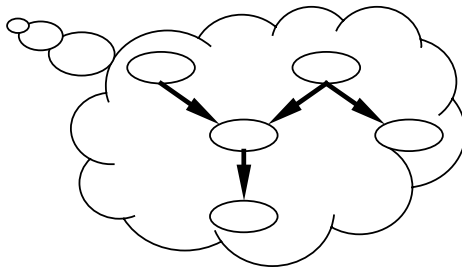
- Conditional independence $\mathbf{X} \perp \mathbf{Y} \mid \mathbf{Z}$

$$P(\mathbf{X} \mid \mathbf{Y}, \mathbf{Z}) = P(\mathbf{X} \mid \mathbf{Z})$$

- Efficient representation of the joint probability distribution using conditional independencies encoded by the graph (network) structure

$$\begin{aligned} P(\mathbf{X}) &= \prod_{i=1}^n P(X_i | X_1, \dots, X_{i-1}) \\ &= \prod_{i=1}^n P(X_i | V(X_i)) \end{aligned}$$

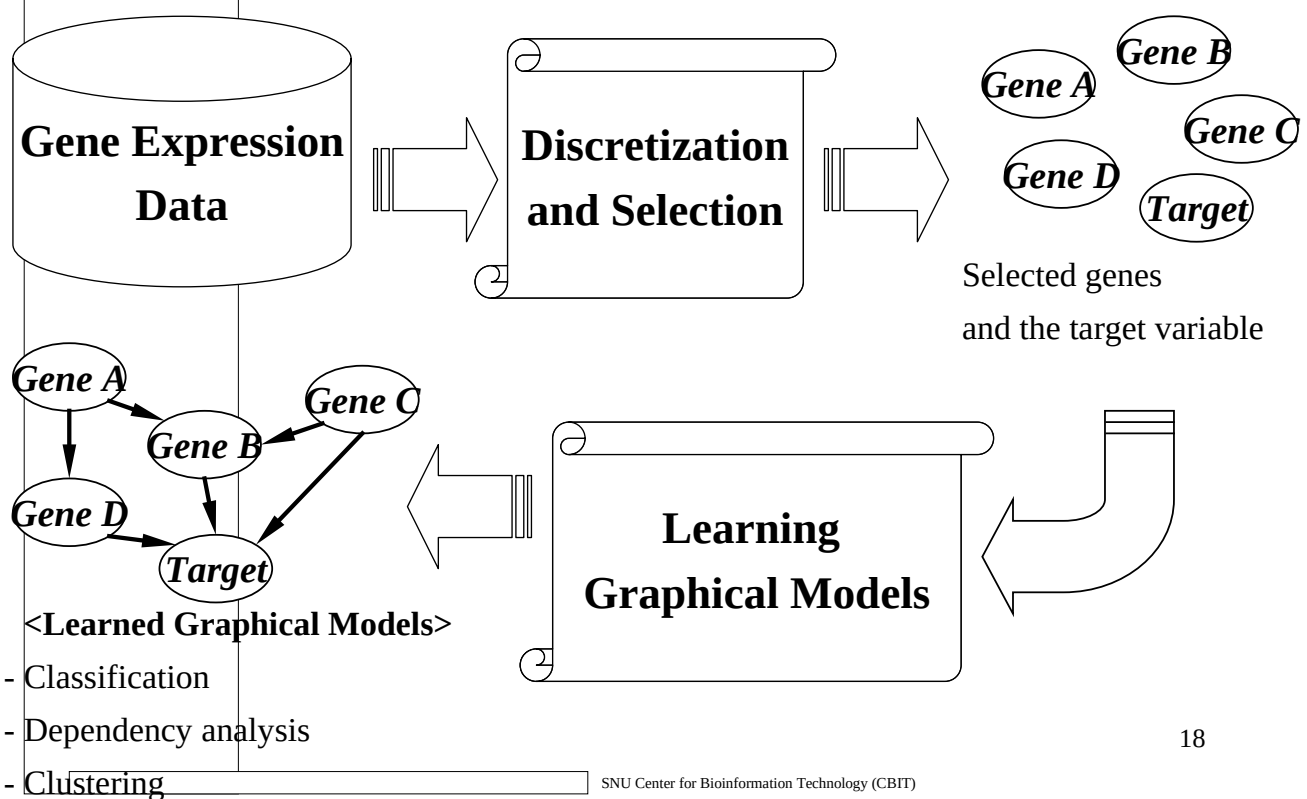
♦ $X_i \perp \{X_1, \dots, X_{i-1}\} - V(X_i) \mid V(X_i)$



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Analysis Procedure



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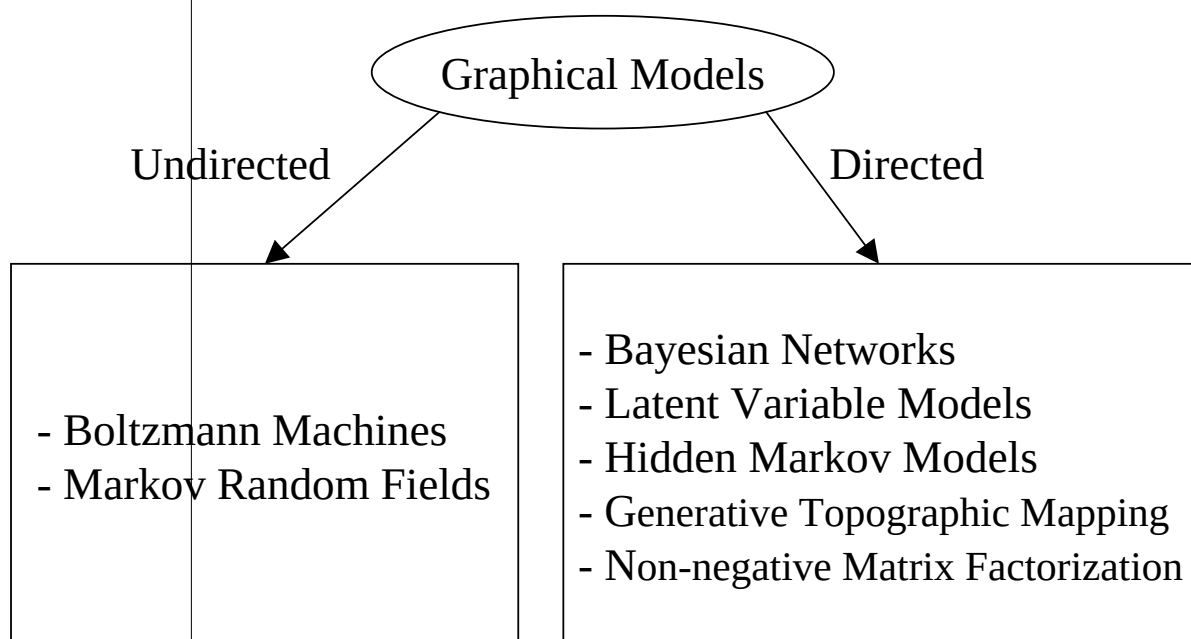
Why Probabilistic Graphical Models?

- The joint probability distribution → all the knowledge about the (biological) system.
- Generative → modeling data sources
- Efficient probabilistic inference → prediction
- The network structure → an insight into the intricate relationships between components (i.e., genes) of the biological system → dependency analysis
- Robust to the noise and error in gene expression data

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Classes of Graphical Models

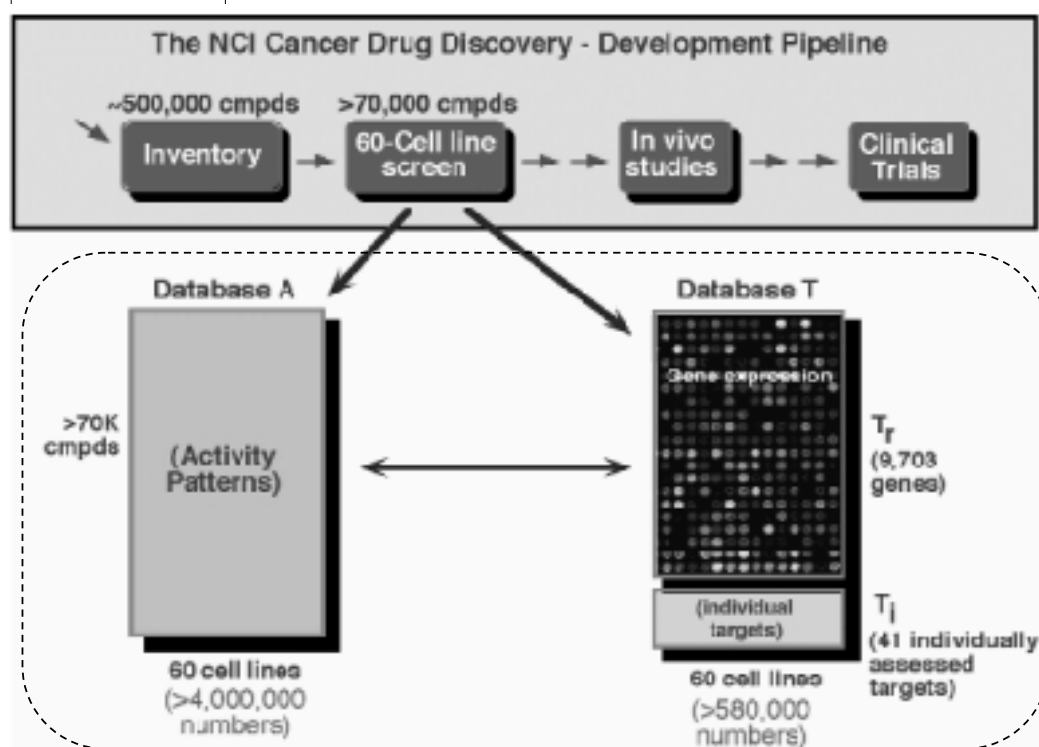


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Gene-Drug Dependency Analysis with Bayesian Networks

NCI Drug Discovery Program



NCI 60 Cell Lines Data Set

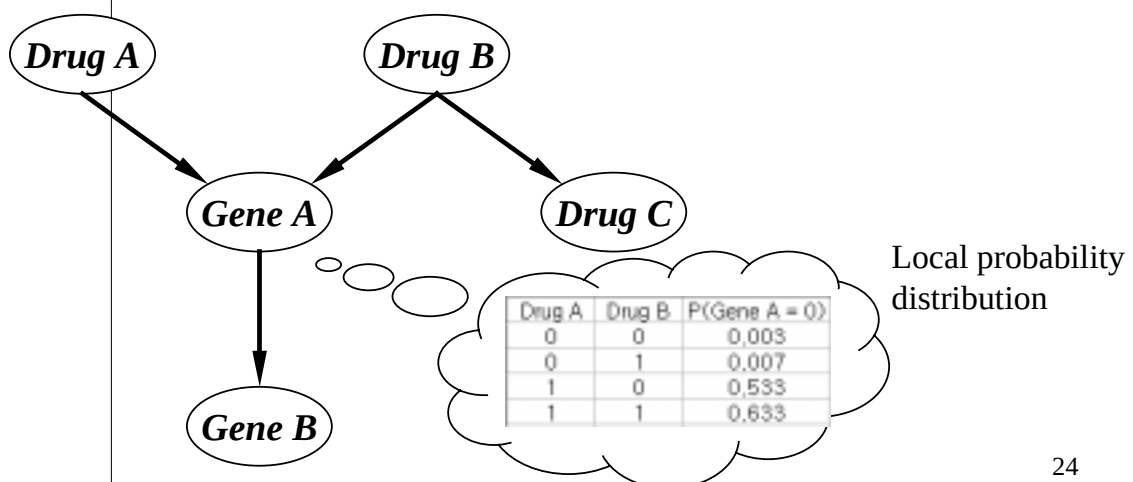
- 60 human cancer cell lines
 - ◆ Colorectal, renal, ovarian, breast, prostate, lung, and central nervous system origin cancers, as well as leukemias and melanomas.
- Drug activity patterns (Database A)
 - ◆ Sulphorhodamine B assay → changes in total cellular protein after 48 hours of drug treatment.
- Individual targets (matrix T_i)
 - ◆ Analysis of molecular characteristics other than mRNA expressions.
- This study focuses on
 - ◆ Sensitivity to therapy $\leftrightarrow$ not molecular consequences of therapy

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Gene-Drug Dependency Analysis Using Bayesian Networks

- To discover
 - ◆ Gene-gene expression dependency
 - ◆ Gene expression-drug activity dependency
 - ◆ Drug-drug activity dependency

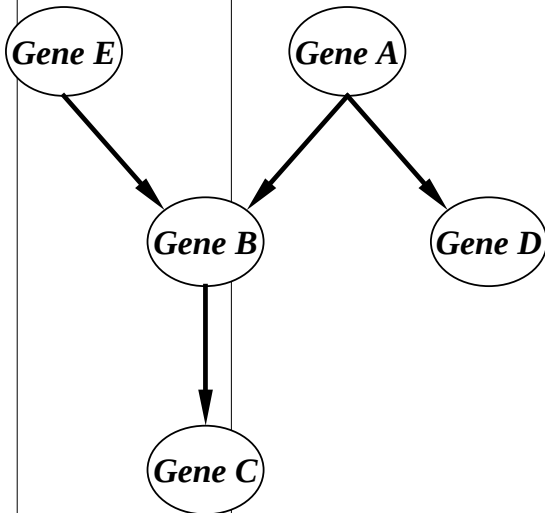


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Bayesian Networks: the Qualitative Part

- The directed-acyclic graph (DAG) structure
 - ◆ Conditional independencies between variables → dependency analysis



<The DAG structure>

<Conditional independencies>

- Gene B and Gene D are independent given Gene A.
- Gene B asserts dependency between Gene A and Gene E.
- Gene A and Gene C are independent given Gene B.

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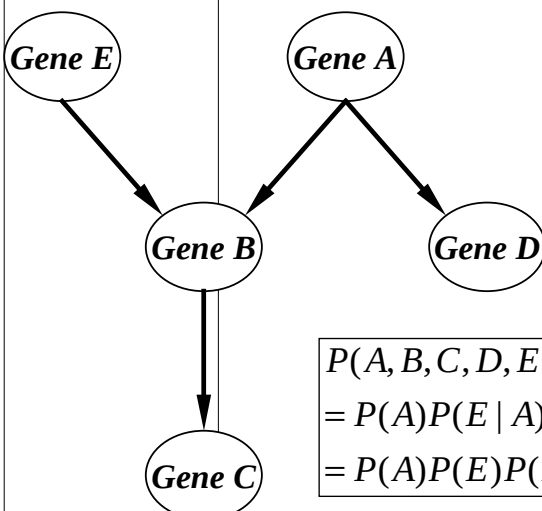
Bayesian Networks: the Quantitative Part

- The joint probability distribution over all the variables in a Bayesian network.

$$P(\mathbf{X}) = \prod_{i=1}^n P(X_i | \mathbf{Pa}_i)$$

Local probability distribution for X_i

$\Theta_i = (\theta_{i1}, \dots, \theta_{iq_i}) \sim$ parameter for $P(X_i | \mathbf{Pa}_i)$
 $P(\theta_{ij}) = \text{Dir}(\theta_{ij} | \alpha_{ij1}, \dots, \alpha_{ijr_i})$
 q_i : # of configurations for $\mathbf{Pa}_i$
 r_i : # of states for X_i



$$\begin{aligned}
 &P(A, B, C, D, E) \\
 &= P(A)P(E | A)P(B | A, E)P(D | A, E, B)P(C | A, E, B, D) \\
 &= P(A)P(E)P(B | A, E)P(D | A)P(C | B)
 \end{aligned}$$

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Applications of Bayesian Networks

- Classification
 - ◆ Bayesian network classifiers
 - ◆ Probabilistic inference → prediction → cancer classification based on gene expression
- Dependency analysis
 - ◆ Bayesian network structure → dependency between components → putative causal relationships

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Discretization of Gene Expression Levels

- The local probability distribution for each variable
 - ◆ Multinomial distribution with Dirichlet prior.

$$p(\theta_{ij} | S^h) = \text{Dir}(\theta_{ij} | \alpha_{ij1}, \alpha_{ij2}, \dots, \alpha_{ijr_i})$$
$$= \frac{\Gamma(\alpha_{ij1} + \dots + \alpha_{ijr_i})}{\prod_{k=1}^{r_i} \Gamma(\alpha_{ijk})} \cdot \prod_{k=1}^{r_i} \theta_k^{\alpha_{ijk} - 1}$$

$$\Gamma(1) = 1, \Gamma(x+1) = x\Gamma(x)$$

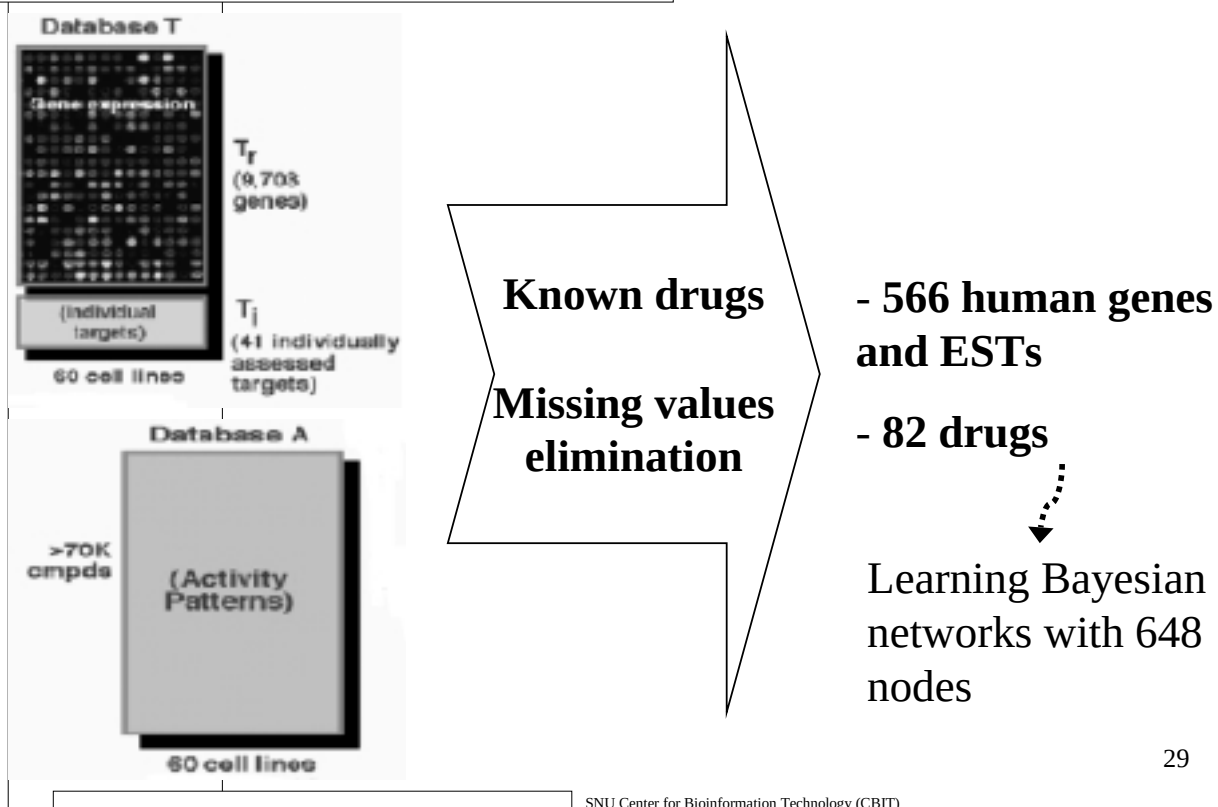
S^h : the network structure

- Discretization of gene expression levels and drug activities
 - ◆ All values → 0 and 1 according to the mean values across the training samples.

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Selection of Genes and Drugs for Analysis



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Learn the Bayesian Network from Data

- Two approaches to Bayesian network learning
 - ◆ Dependency analysis-based approach
 - ◆ Optimization-based approach
 - Network score: fitness of the network to training data D .
 - Search the best-scoring network structure
- Scoring metric for the network
 - ◆ MDL (minimum description length) score
 - ◆ BD (Bayesian Dirichlet) score

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BD (Bayesian Dirichlet) Score

- Assumptions

- ◆ Multinomial sample
- ◆ Parameter independence and modularity
- ◆ Dirichlet prior

$$p(\theta_{ij} | S^h) = c \cdot \prod_{k=1}^{r_i} \theta_{ijk}^{\alpha_{ijk} - 1}$$

- ◆ Complete data

- BD score is calculated as follows:

$$p(D, S) = p(S) \cdot p(D | S)$$

$$= p(S) \cdot \prod_{i=1}^n \prod_{j=1}^{q_i} \frac{\Gamma(\alpha_{ij})}{\Gamma(\alpha_{ij} + N_{ij})} \prod_{k=1}^{r_i} \frac{\Gamma(\alpha_{ijk} + N_{ijk})}{\Gamma(\alpha_{ijk})}$$

Prior probability

Sufficient statistics
calculated from D

- ◆ S : the network structure, D : the training data

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Search Strategy

- Finding the best network structure → NP-hard problem.
 - ◆ Exponential time complexity.
 - ◆ In general, greedy search or simulated annealing is used.
- General greedy search algorithm is not applicable to the construction of large-scale Bayesian networks with hundreds of nodes.
 - ◆ Time and space complexity
- Reduce the search space
 - ◆ “Sparse candidate algorithm” by [Friedman et al. 1999]
 - ◆ “Local to global search algorithm”
 - Exploit the local search for the Markov blanket of each node to reduce the global search space.

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Local to Global Search Algorithm

Input:

- A data set D .
- An initial Bayesian network structure B_0 .
- A decomposable scoring metric,

$$\text{Score}(B, D) = \sum_i \text{Score}(X_i \mid \text{Pa}^B(X_i), D).$$

Output: A Bayesian network structure B .

Loop for $n = 1, 2, \dots$, until convergence.

-Local Search Step:

* Based on D and B_{n-1} , select for each variable X_i , a set $CB_i^n (|CB_i^n| \leq k)$ of candidate Markov blanket of X_i .

* For each set $\{X_i, CB_i^n\}$, learn its local structure and determine the Markov blanket of X_i , $BL^n(X_i)$, from this local structure.

* Merge all the local network structures $\mathcal{G}\{X_i, BL^n(X_i)\}, E_i$ into a global network structure H_n (usually cyclic).

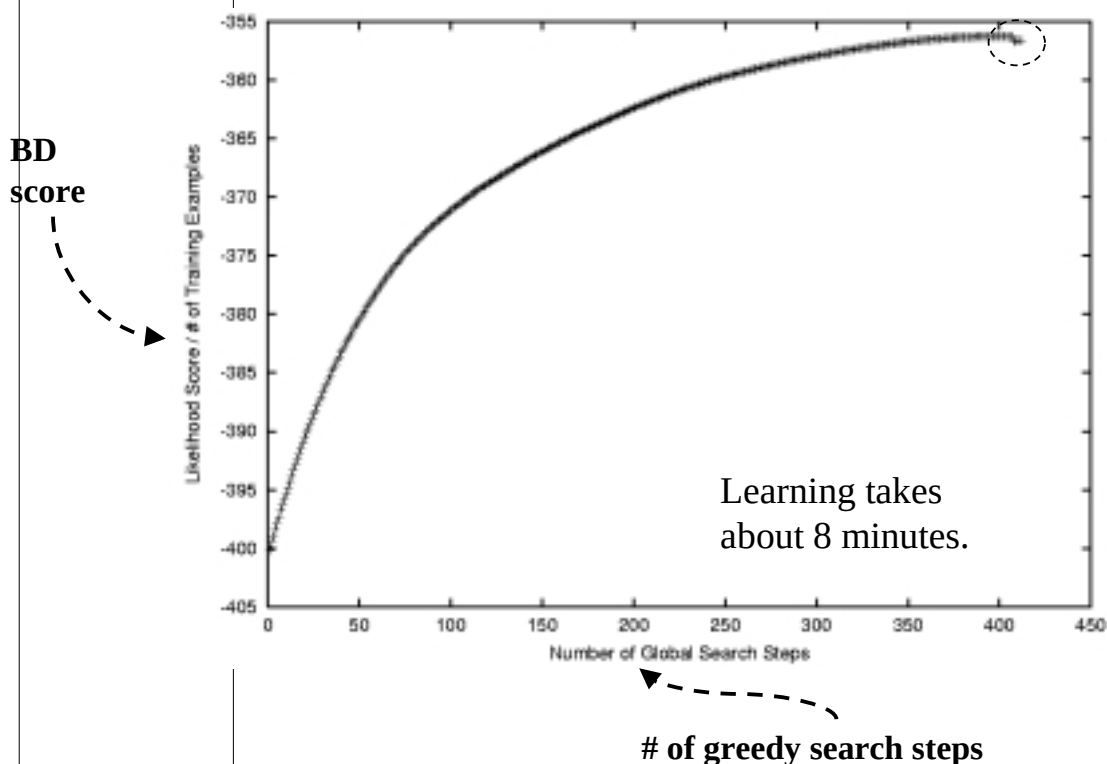
-Global Search Step:

* Find the Bayesian network structure $B_n \subset H_n$, which maximizes $\text{Score}(B_n, D)$ and retains all non-cyclic edges in H_n .

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Learning Curve



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Drug-Drug Dependency

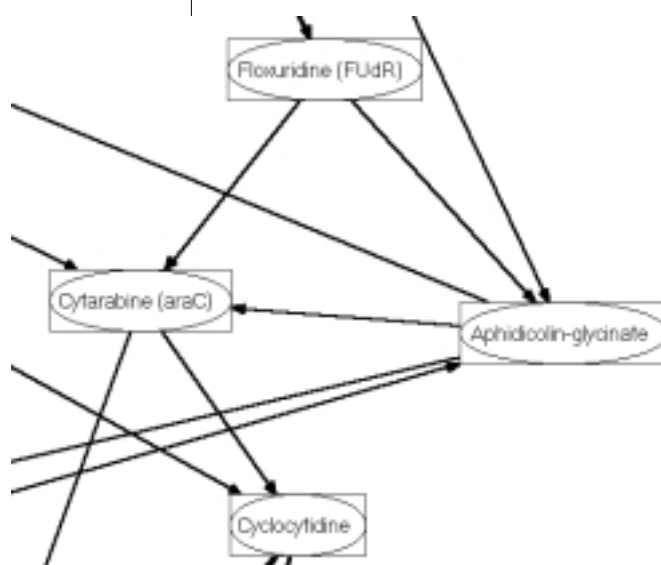
- The pyrimidine analogues
 - ◆ Aphidicolin-glycinate
 - ◆ Floxuridine
 - ◆ Cyclocytidine
 - ◆ Cytarabine
- ⇒ DNA synthesis inhibitor
- Shows similar activity pattern across 60 cancer cell lines
 - Clustered together in [Scherf et al. 2000].

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Experimental Results: Drug-Drug Dependency

- Drug-drug activity correlations



- Three drugs “Aphidicolin-glycinate”, “Floxuridine”, and “Cytarabine” directly depend on each other.

- “Cyclocytidine” directly depends on “Cytarabine” and vice versa.

⇒ **Confirmation**

<Part of the learned Bayesian network structure>

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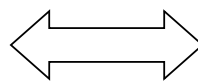
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Gene-Drug Dependency

- Gene “ASNS”

- ◆ Certain malignant cells, including many acute lymphoblastic leukemias (ALL) lack asparagine synthetase (ASNS) → exogenous “L-asparagine” [Scherf et al. 2000].

Sensitivity of the drug
“L-asparagine”



Expression levels of the
gene “ASNS”

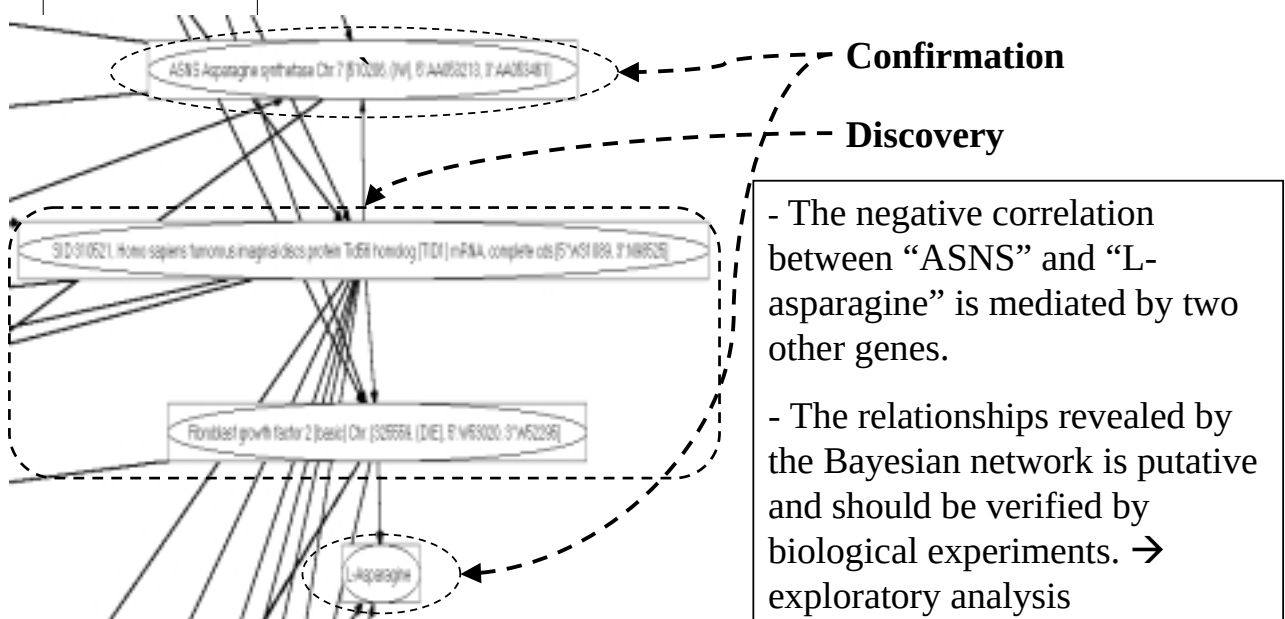
High negative correlation

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Experimental Results: Gene-Drug Dependency

- Gene expression-drug activity correlations

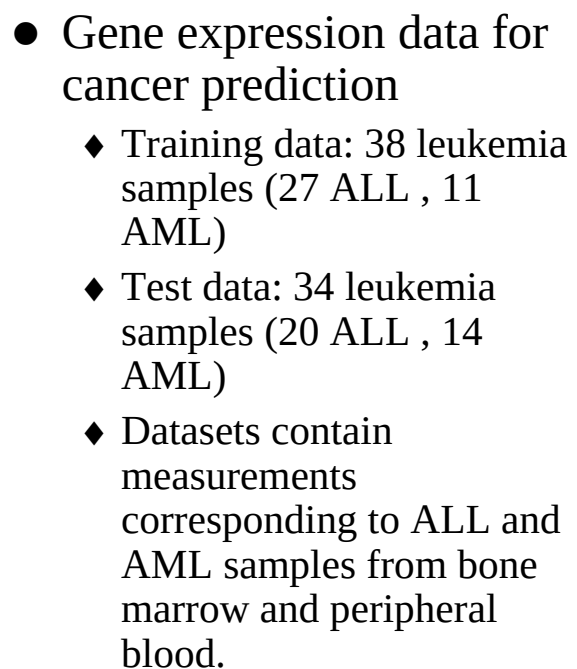


<Part of the learned Bayesian network structure>

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CAMDA-2000 Data Set



Cluster Analysis Using Non-negative Matrix Factorization

- Method
 - ◆ Using NMF for class clustering and prediction of gene expression data from acute leukemia patients
- NMF (non-negative matrix factorization)

$$\mathbf{G} \approx \mathbf{WH}$$

$$(\mathbf{G})_{i\mu} \approx (\mathbf{WH})_{i\mu} = \sum_{a=1}^r W_{ia} H_{a\mu}$$

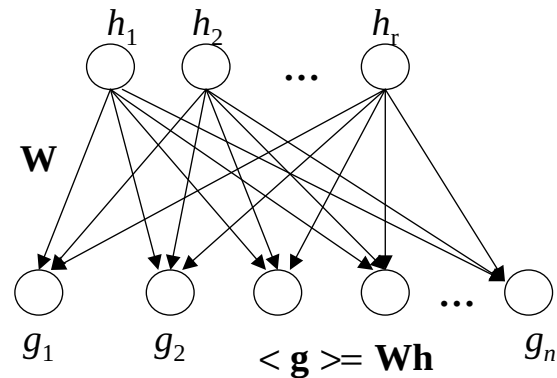
$\mathbf{G}$: gene expression data matrix

$\mathbf{W}$: basis matrix (prototypes)

$\mathbf{H}$: encoding matrix (in low dimension)

$$G_{i\mu}, W_{ia}, H_{a\mu} \geq 0$$

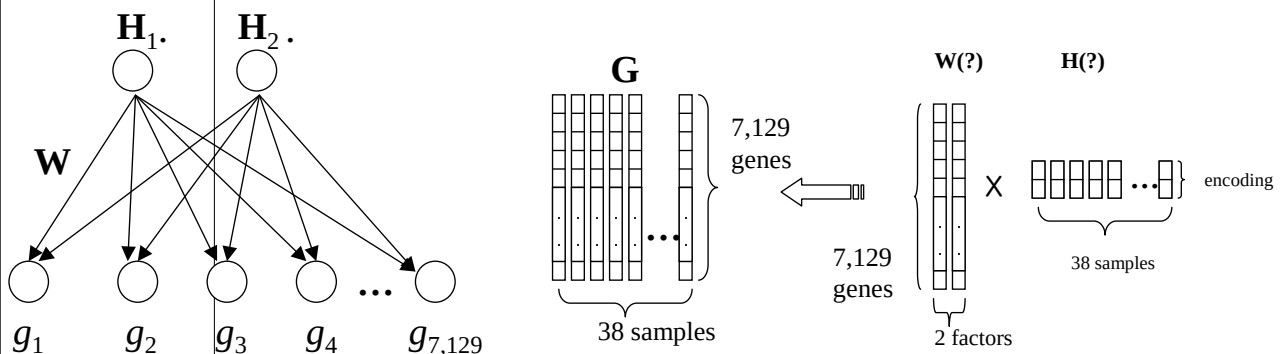
- NMF as a latent variable model



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Clustering Gene Expression Data



- Factors can capture the correlations between the genes using the values of expression level.
- Cluster training samples into 2 groups by NMF
 - ◆ Assign each sample to the factor (class) which has higher encoding value.

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Learning Procedure

Input : Gene expression data matrix, $\mathbf{G}$ ($n \times m$)

Output : base matrix $\mathbf{W}$ ($n \times k$), encoding matrix $\mathbf{H}$ ($k \times m$)

n : data size, m : number of genes, k : number of latent variables

Objective function :
$$F = \sum_{i=1}^n \sum_{\mu=1}^m [G_{i\mu} \log(WH)_{i\mu} - (WH)_{i\mu}]$$

Procedure

1. Initialize $\mathbf{W}$, $\mathbf{H}$ with random numbers.

$$W_{ij} \geq 0, \sum_j W_{ij} = 1$$

$$H_{ij} \geq 0$$

2. Update $\mathbf{W}$, $\mathbf{H}$ iteratively until max_iteration or some criterion is met.

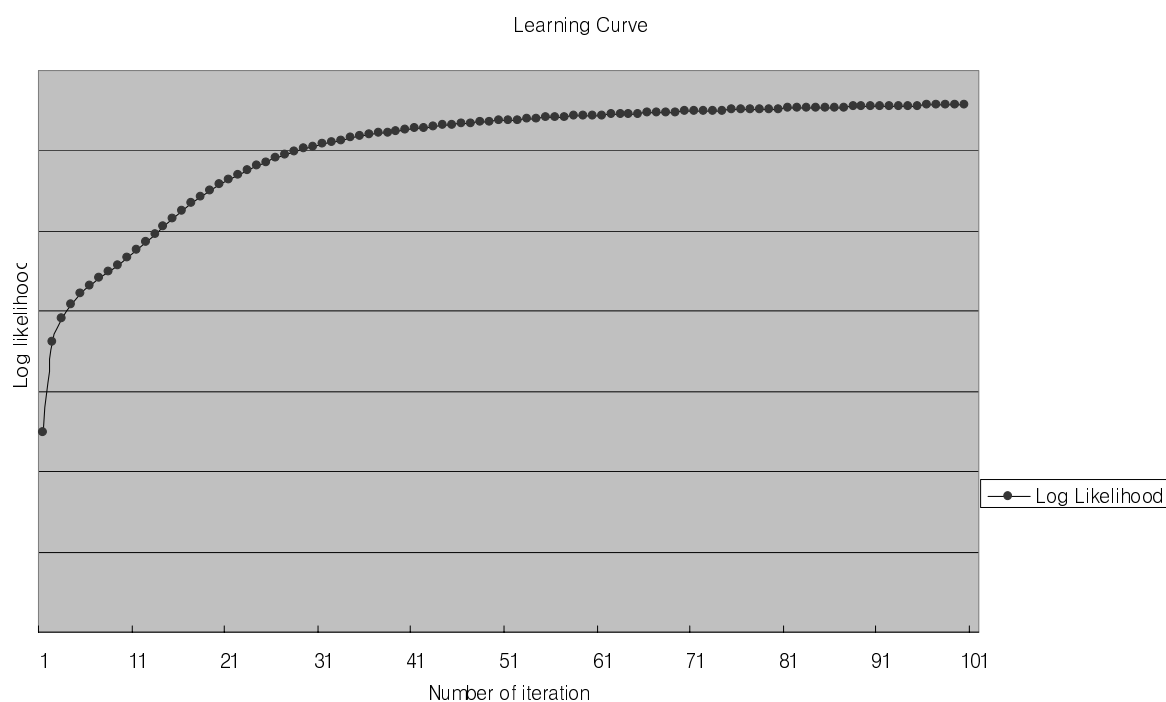
$$W_{ia} \leftarrow W_{ia} \sum_{\mu} \frac{G_{i\mu}}{(WH)_{i\mu}} H_{a\mu} \quad H_{a\mu} = H_{a\mu} \sum_i W_{ia} \frac{G_{i\mu}}{(WH)_{i\mu}}$$

$$W_{ia} \leftarrow \frac{W_{ia}}{\sum_j W_{ja}}$$

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Learning Curve

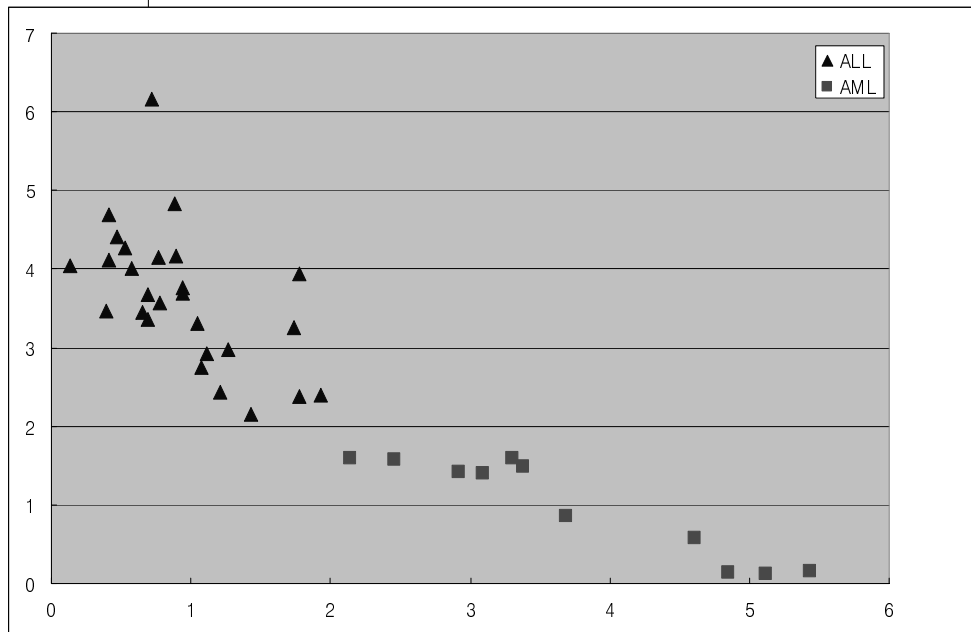


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Experimental Results: Cancer Clustering

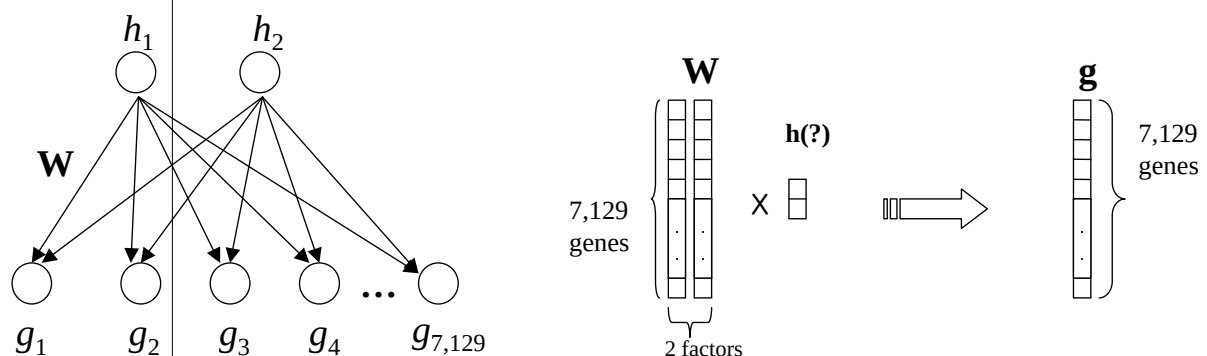
Accuracy: 0 ~1 error for the training data set



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Diagnosis Using NMF



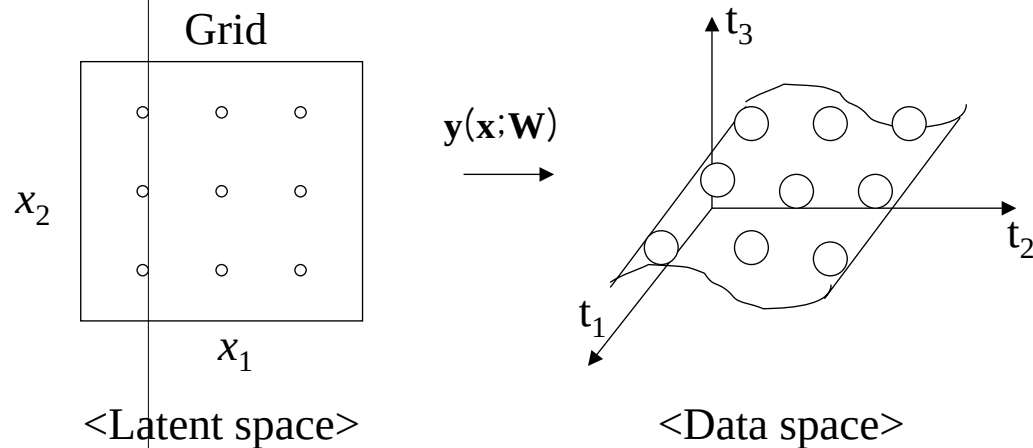
- For each test sample g , estimate the encoding vector h that best approximates the sample.
 - ◆ W is the basis matrix computed during training (fixed).
 - ◆ As in training, assign each sample to the factor (class) which has the highest encoding value.
- Accuracy: 1~2 error(s) for the test data set

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Clustering Using Generative Topographic Mapping

- GTM: a nonlinear, parametric mapping $y(\mathbf{x}; \mathbf{W})$ from a latent space to a data space.



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Learning Generative Topographic Mapping

- Learning algorithm
 - Generate the grid of latent points.
 - Generate the grid of latent function centers.
 - Compute the matrix of basis function activations Φ .
 - Initialize weights $\mathbf{W}$ in $\mathbf{Y} = \Phi \mathbf{W}$ and the noise variance β .
 - Compute $\Delta_{n,k} = \|\mathbf{t}_n - \Phi_k \mathbf{W}\|^2 = \|\mathbf{t}_n - \mathbf{y}_k(\mathbf{x}, \mathbf{W})\|^2$ for each n, k
 - Repeat
 - ▶ - Compute the responsibility matrix $\mathbf{R}$ using Δ and β . [E-Step]
 - ▶ Compute $\mathbf{G} = \mathbf{R}^T \mathbf{R}$
 - ▶ - Update $\mathbf{W}$ by $\Phi^T \mathbf{G} \Phi \mathbf{W} = \Phi^T \mathbf{R} \mathbf{T}$ [M-step]
 - ▶ - Compute $\Delta = \|\mathbf{t} - \Phi \mathbf{W}\|^2$
 - ▶ - Update β
 - Until convergence

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GTM: Visualization

- Posterior distribution in latent space given a data point $\mathbf{t}$:

$$\mathbf{X}(\mathbf{t}) \sim$$

$$p(\mathbf{x}_k|\mathbf{t}) = \frac{p(\mathbf{t}|\mathbf{x}_k, \mathbf{W}^*, \beta^*)p(\mathbf{x}_k)}{\sum_{k'} p(\mathbf{t}_n|\mathbf{x}_{k'}, \mathbf{W}^*, \beta^*)p(\mathbf{x}_{k'})}.$$

- For a whole set of data, for each $\mathbf{t}$, plot in the latent space
 - ◆ Posterior mode:

$$\mathbf{x}_n^{\text{mode}} = \underset{\mathbf{x}_k}{\operatorname{argmax}} p(\mathbf{x}_k|\mathbf{t}_n),$$

- ◆ Posterior mean:

$$\mathbf{x}_n^{\text{mean}} = \sum_k^K \mathbf{x}_k p(\mathbf{x}_k|\mathbf{t}_n).$$

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GTM: Clustering Experiment

- Gene Selection
 - ◆ Select about 50 genes out of 7,129 based on the three test scores of cancer diagnosis.
 - Correlation metric (similar as t -test)
 - Wilcoxon test scores (a nonparametric t -test)
 - Median test scores (a nonparametric t -test)
- Clustering & Visualization
 - ◆ After learning a model, genes are plotted in the latent space.
 - ◆ With the mapping in the latent space, clusters can be identified.

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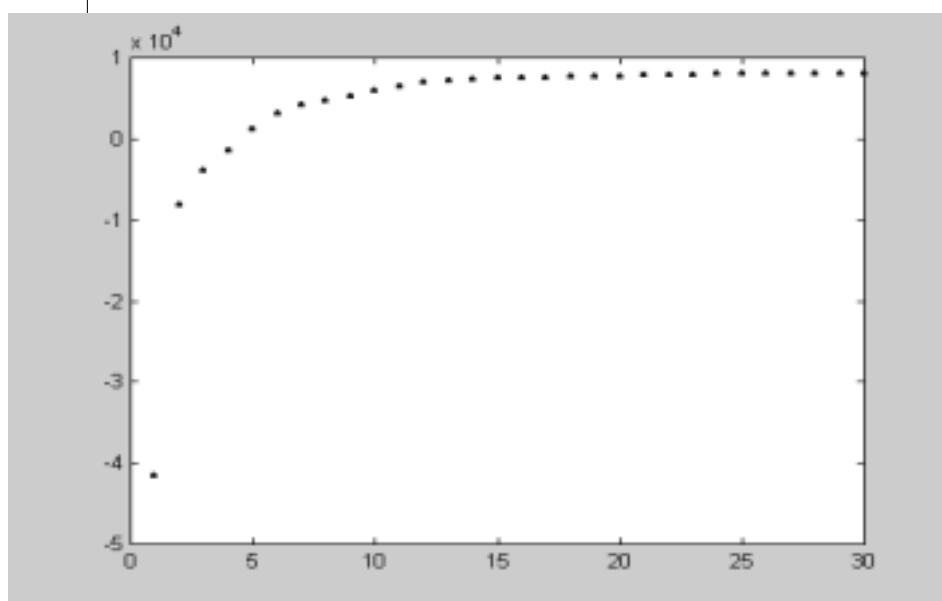
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List of Genes Selected

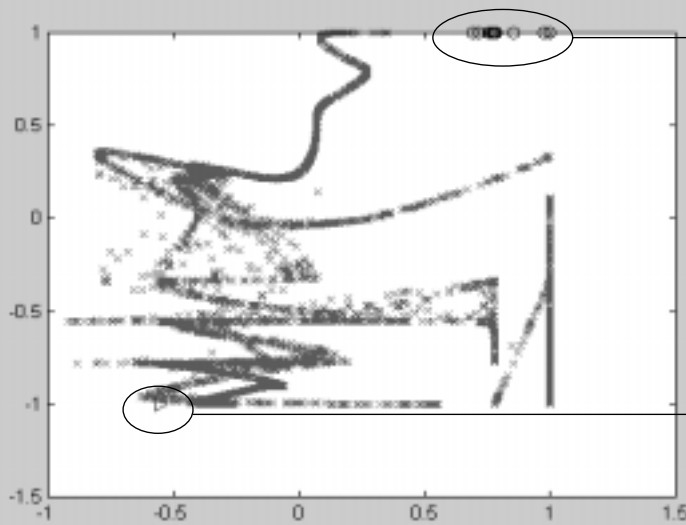
Gene Description	Gene Accession Number	x1(Normal)	x37(Cancer)	P2_WIL	P2_MED	P-Means
PAH Fumarylacetoacetate	M55150_at	654	2072	0.00001	0.00001	-1.4676
Leukotriene C4 synthase (LTC4S) gene	U50136_ma1_at	1124	2539	0	0.00001	-1.4217
Zyxin	X55735_at	290	2612	0	0.00001	-1.4057
LYN Y=ves-1 Yarnasuchi sarcoma viral related oncogene homolog	M16030_at	365	2399	0.00001	0.00001	-1.2029
CD33 CD33 antigen (differentiation antigen)	M23197_at	261	400	0	0.00001	-1.1959
DF D component of complement (adipsin)	M04526_at	-283	6799	0.00008	0.00149	-1.1910
LEPR Leptin receptor	Y12670_at	600	1064	0.00001	0.00001	-1.1847
GB DEF = Homeodomain protein HoxA9 mRNA	U82758_at	393	805	0.00003	0.00149	-1.1641
Uter mRNA for interferon-gamma inducing factor(IGIF)	D48650_at	75	311	0.00009	0.01348	-1.1597
CST3 Cystatin C (amyloid angiopathy and cerebral hemorrhage)	M27691_at	303	13296	0	0.00001	-1.1091
PRG1 Proteolysin 1, secretory granule	X17042_at	177	8479	0.00004	0.00149	-1.1059
Thrombospondin-p50 gene extracted from Human thrombospondin-1 gene, partial c	U12471_cds1_at	180	417	0.00004	0.00001	-1.0697
Phosphotyrosine independent ligand p82 for the Lck SH2 domain mRNA	U46751_at	1298	6903	0.00085	0.01348	-1.0640
INTERLEUKIN-8 PRECURSOR	Y50787_s_at	333	19361	0.00003	0.00149	-1.0409
INDUCED MYELOID LEUKEMIA CELL DIFFERENTIATION PROTEIN MCL1	L88246_at	543	5503	0.00006	0.00149	-1.0347
PEPTIDYL-PROLYL CIS-TRANS ISOMERASE, MITOCHONDRIAL PRECURSOR	M80254_at	-88	672	0.00008	0.00149	-1.0277
ATP6C Vacuolar H+ ATPase proton channel subunit	M62762_at	835	3808	0.00007	0.00149	-1.0228
CDC25A Cell division cycle 25A	M81933_at	-240	330	0.00001	0.00001	-1.0061
Azurocidin gene	M96326_ma1_at	782	9933	0.00006	0.00149	-1.0053
Transcriptional activator hSNF2b	U29175_at	1582	393	0.00011	0.00001	0.9060
ACADM Acyl-Coenzyme A dehydrogenase, C-4 to C-12 straight chain	M91432_at	767	183	0.00008	0.00001	0.9135
HKR-T1	S50223_at	268	-53	0.00017	0.00001	0.9215
Clone 22 mRNA, alternative splice variant alpha:1	AF009426_at	36	-45	0.00009	0.00001	0.9246
IRF2 Interferon regulatory factor 2	X15946_at	277	14	0.00045	0.00007	0.9307
CTPS CTP synthetase	X52142_at	107	-84	0.00014	0.00001	0.9331
CTP28 CTP synthetase (DNA) II beta (180kD)	Z15115_at	9444	802	0.00005	0.00001	0.945
CD19 CD19 antigen	M28170_at	397	-21	0.00028	0.00001	0.9494
Inducible protein mRNA	L47796_at	571	-80	0.00002	0.00001	0.9504
Cytoplasmic dynein light chain 1 (hdlg1) mRNA	U32944_at	3349	224	0.00051	0.00001	0.9548
TCF3 Transcription factor 3 (E2A immunoglobulin enhancer binding factors E12/E47)	M31523_at	1320	190	0.00001	0.00001	0.9712
Transcriptional activator hSNF2b	D28156_s_at	1995	697	0.00009	0.00001	0.989
Thymopiesin beta mRNA	U99087_s_at	358	52	0.00014	0.00001	1.0142
MYL1 Myosin light chain (alkali)	M31211_s_at	601	139	0.00002	0.00001	1.0227
CRY2 Crystallin zeta (quinone reductase)	L13278_at	193	-32	0.00016	0.00001	1.0237
RETINOBLASTOMA BINDING PROTEIN P48	X74282_at	1372	130	0.00002	0.00001	1.0277
CCND3 Cyclin D3	M92287_at	4778	852	0.00001	0.00001	1.0430
ME-1 gene	U55259_ma1_at	9326	218	0.00028	0.00001	1.1051
PROTEASOME XOTA CHAIN	X59417_at	3016	528	0.00004	0.00001	1.1246
C-myb gene extracted from Human (c-myb) gene, complete primary cds, and five c	U22375_cds2_s_at	3105	186	0.00001	0.00001	1.5593

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GTM: Learning Curve



GTM: Clustering Result



Genes with high expression levels in case of ALL (large P-metric value)

Genes with high expression levels in case of AML (negative large P-metric value)

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Experimental Results: Clusters Found by GTM

- Three cell cycle-regulated clusters found by GTM [Shin et al. 2000]

	Cluster center	No. of train Data/ no. in cluster	Correct no. / test data	Overall mean expression levels (Cln/b) of known genes
S/G2		5 /	1 / 2	(.148 .184 -.367 -.044)
S	(0.111 -0.333)	5 / 5	5 / 5 (100%)	(1.075 1.482 -.233 -.375)
M/G1 c1	(0.111 0.333)	13 / 7	1 / 6	(-.171 -.573 .091 .311)
c2	(-0.111 -0.111)	/ 2	0 / 6	
c3	(0.323 0.1)	/ 2	0 / 6	
G2/M c1	(0.111 0.333)	10 / 5	0 / 5	(-.616 -1.01 1.832 1.596)
c2	(0.111 0.111)	/ 3	3 / 5 (80%)	
G1 c1	(-0.111 0.333)	35 / 18	10 / 16 (62%)	(.894 .907 -.766 -.479)
c2	(-0.111 0.111)	/ 7	0 / 16	

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Experimental Results: Comparison with Other Methods

● Comparison of prototype expression levels

	No. of selected genes	Mean expression levels by GTM	No. of selected genes by Spellman	Mean expression levels by Spellman
S/G2	92	(.13 -.06 -.1 .01)	121	(.13 .05 -.16 .03)
S	25	(.84 .81 -.42 -.33)	71	(.46 .47 -.43 -.18)
M/G1 c1	120	(.82 .65 -.65 -.38)	113	(-.21 -.61 -.04 .07)
c2	34	(-.04 -.37 -.01 -.11)		
c3	10	(.32 .29 -.3 .05)		
G2/M c1	33	(-.59 -.96 1.34 1.29)	195	(-.32 -.62 .49 .54)
c2	60	(.08 -.30 .51 .57)		
G1 c1	122	(.92 .74 -.62 -.33)	300	(.66 .49 -.55 -.33)
c2	74 (total = 570)	(.79 .82 -.48 -.34)		
			(total = 800)	

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Conclusion

Summary and Future Work

- A class of nonstandard learning algorithms, i.e. probabilistic graphical models, is presented for microarray data analysis:
 - ◆ Bayesian networks
 - ◆ Non-negative matrix factorization
 - ◆ Generative topographic mapping
- Probabilistic graphical models are useful for biological data mining:
 - ◆ Comprehensibility
 - ◆ Generative
 - ◆ Soft inference (clustering)
 - ◆ Dependency analysis
- Future work includes:
 - ◆ Construction of larger-scale networks
 - ◆ Acceleration of learning processes
 - ◆ Handling data sparseness and missing data problems

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